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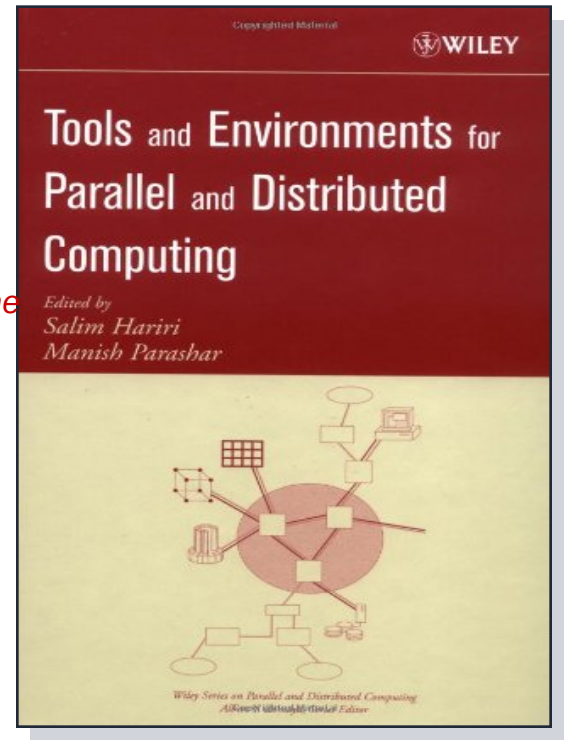

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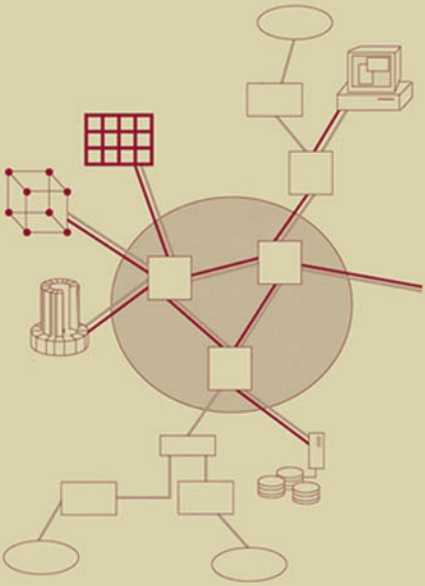

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PREFACE

The primary focus of this book is the rapidly evolving software technology for supporting the development, execution, management, and experimentation with parallel and distributed computing environments. The design, development, and utilization of parallel and distributed computing environments that can efficiently support a wide range of scientific and engineering applications remains a challenging research problem due to the complexity and varying requirements of the applications, heterogeneity of the computing systems and their networks, asynchronous complex interactions among the system and application components, and the heterogeneity of the software tools and environments. However, recent advances in processing and network technology and software tools have addressed successfully many of the obstacles hindering the wide deployment of parallel and distributed computing environments.

Active research in parallel processing has resulted in advances in all aspects of the computing technologies, including processing technology, computer networking technology, and software technology. Advances in processing technology have resulted in faster, more powerful processors with increased functionality. Advances in computer networking technology have introduced reliable high-speed networks capable of providing high transfer rates. Advances in software technology have provided easy-to use tools and environments for the development of parallel applications. These advances have resulted in the proliferation of a large number of different architectural classes, such as SIMD computers, MIMD computers, vector computers, and data flow computers, where each class represents a set of different trade-offs in design decisions such as coarse-grained (MIMD) parallelism versus fine-grained (SIMD) parallelism, shared memory MIMD versus distributed memory MIMD, hypercube topology versus mesh topology, and circuit-switched versus packet-switched communication. Each architectural class is tuned to deliver maximum performance to a specific set of applications. However, it remains a fact that none of the existing computing systems are general enough to address all classes of applications and provide the desired performance levels. In addition, these architectures are not scalable and their relatively narrow applicability has prevented them from being cost-effective.

Furthermore, the development of efficient application software capable of exploiting the available computing potential is nontrivial and requires a thor-

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